

# Apoorva Sharma

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[Gmail](#) . [GitHub](#) . [LinkedIn](#)

## TECHNICAL SKILLS

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**Programming Languages:** Python, C++, SQL

**AI/ML:** PyTorch, CNNs, Transformers, YOLO, OpenCV, NLP, Deep Learning

**Frameworks & Tools:** FastAPI, Flask, Firebase, Streamlit, Git/GitHub

## PROJECTS

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### 1. Multimodal Emotion Recognition System — *PyTorch, FastAPI, OpenCV, YOLO, Librosa* — [GitHub](#)

- Engineered a real-time multimodal emotion recognition pipeline integrating facial, speech, and text modalities using ResNet18, LSTM, and DistilBERT models trained on FER2013, RAVDESS, and GoEmotions datasets.
- Designed confidence-aware late fusion mechanism combining modality-level probability distributions with adaptive weight boosting for high-confidence predictions.
- Built FastAPI inference backend supporting webcam/image, audio, and text inputs with modular training and inference pipelines.
- Achieved 82% text classification accuracy using DistilBERT and 61% multimodal visual/audio classification accuracy under real-time inference constraints.

### 2. AI-Based Autonomous Robot for Face Recognition & Navigation — *Python, OpenCV, Raspberry Pi, Arduino* — [GitHub](#)

- Developed an AI-assisted autonomous robotic system for real-time face recognition and navigation using OpenCV, Raspberry Pi, and Arduino.
- Implemented hardware-software communication pipelines for responsive motion control and environment-aware robotic behavior.
- Worked on edge-based computer vision and navigation workflows during research internship at Bhabha Atomic Research Centre.

### 3. ManoSamvada – AI Mental Wellness Chatbot — *Flask, MySQL, Groq API, Python* — [GitHub](#)

- Developed an AI-powered mental wellness chatbot using Flask, MySQL, and Groq LLM APIs for emotion-aware conversational support.
- Designed backend APIs for authentication, session management, reporting workflows, and emotional analytics dashboards.
- Implemented crisis keyword detection and conversational tracking pipelines with modular chatbot service architecture.

## EXPERIENCE

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**Research Intern** — Bhabha Atomic Research Centre . [Certificate](#)

Jul 2024 – Nov 2024

- Worked on AI-based autonomous robotics using OpenCV, Raspberry Pi, and Arduino.
- Assisted in face recognition, navigation, and hardware-software integration workflows.

## EDUCATION

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**M.Tech Artificial Intelligence** — Banasthali Vidyapith — 2025–Present — CGPA: 7.8

**B.Tech Computer Science Engineering** — Banasthali Vidyapith — 2021–2025 — CGPA: 7.65